

Jun Tan

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EDUCATION

MSc Applied Machine Learning, Imperial College London, UK Sep 2025 – Current

Dissertation: Task-based Transfer RL (TBTRL) with F-N-B factorisation. While high-dimensional RL problems face sample efficiency and long training time issues, knowledge reuse is crucial across tasks. Building a new RL algorithm that learns representation via Contrastive RL and a custom FNB method, achieving few-shot or zero-shot performance when a new task is given, independent of the reward function, on top of frontier papers on ExORL and OGBench. (PyTorch, Mujoco, DRL, SAC, TD3, PPO, FB, TD-JEPA)

ML Systems Highlights: LoRA fine-tuning of BERT using mase tools, Paper review on Convergence Performance of Split-Federated Learning in a heterogeneous setting, NAS and precision search for DNN using Optuna, SDPA kernel-fusion, MXINT format quantisation, writing CUDA kernel for thread and data tiling, Optimisation and Distributed Optimisation via graphs and a decentralised setting.

BEng Mechatronics Engineering with Industrial Studies, University of Southampton, UK Oct 2021 – Jun 2024

First Class Honours: 80.9% | 2022 Zepler Prize | 2024 G-Research Prize - Top of the Class

Dissertation: Q-Learning Controlled (Q-LEAC) Stroke Rehabilitation System. Invented a MATLAB-C++ RL control system for stroke rehabilitation using Q-Learning for forearm muscle stimulation and UltraLeap camera vision feedback for a real-time deployment system, surpassing traditional model-based control methods' live training time by more than 50% and tested on patients in the hospital.

WORK EXPERIENCE

Mercedes-AMG High-Performance Powetrains (HPP) Sep 2024 – Sep 2025

Software and Controls Placement - SECU, Lap Optimisation, and Software Reliability Team, F1 R&D

- 2026 F1 Power Unit Project - Spearheaded the development and maintenance of the on-car software systems, ensuring the ECU maintains good thread balancing and processor loadings across all cores to maximise computational efficiency of the engine on track.
- Worked closely with the Lap Simulation team in Energy Deployment Optimisers to extract lap time gains by improving convergence speed and refining constrained optimisation algorithms under complex regulations from FIA.
- Implemented low-level code logic in Simulink for F1 on-car systems, including software architecture, engine operations, chassis and brake control, with SiL and HiL verification via System Monitor/ATLAS based on Root Cause Analysis and fault/bug data investigation.
- Delivered software requests from multiple departments using Git, Azure DevOps for pipeline management and software releases, while developed MATLAB/YAML automation scripts to accelerate the code building process and take over repetitive tasks.

Allye Energy (Startup) - Energy Storage and Engineering Intern Jun 2024 – Aug 2024

- Delivered full-stack HMI software development with a dashboard for energy harvest/deployment, SOC, and data visualisation upon connection with BESS units, and a backend API server fetching JSON data via localised MongoDB, using TypeScript and Java.
- Maintained and analysed an ML prediction model for electrical harvest/deployment to optimise efficiency and balance load distribution with MAX300 BESS unit (Built from 4 Tesla EV batteries as power supply), reducing the cost of grid energy harvesting by up to 24%.

Dyson - Mechatronics Engineering Intern - Measurement and Capability Team Jun 2023 – Sep 2023

- Developed modular and sustainable LabVIEW code for an automated brushbar test rig that refactored and reduced the computational cost by 40%, alongside the design of 3D printed adapters via NX12 for product compatibility across the business.
- Established the habit of using Bitbucket for source control, while producing readable codes in OOP style upon PRs through cross-team collaboration. Also documented issues and changes with Confluence and Jira for traceability and agile development process.

PROJECTS

Glove-based Invisible Keyboard (GIK) Imperial26 - Built a pair of gloves controlled by NanoRev2, IMUs and FSRs to capture finger and hand movements and achieve keyboard-free typing on any surface. The custom REFFCO Deep ML architecture, which uses attention LSTM, LM fusion, and Optuna takes in 118-dimensional data and predicts the characters upon surface contact during live inference.

Mase_EDGE Imperial26 - Built a hardware-in-the-loop optimisation pipeline that automates the adaptation of Deep NN model for efficient execution on Android devices. The pipeline uses Optuna to search for optimal hyperparameters, including backend delegation ratio of CPU-GPU-NPU, followed by quantisation (QAT/PTQ) and pruning upon running the app on the device. It makes decisions based on the model's real-time feedback within the study, achieving hardware-aware deployment tailored to user's Android device specifications.

Autonomous Rover Project Soton23 - Achieved the highest mark ever of the module by design and built a self-navigating rover with path-planning ability (SLAM) that used LIDAR and gyro sensors to provide real-time feedback and control of the environment.

OpenAI Parameter Golf Challenge Optimised LM model with XSA, MXINT4 based QAT, Muon optimiser and unstructured pruning.

HOBBIES AND ACHIEVEMENTS

- **Malaysia National Speedcuber** – Rubik's WCA World Championship 2023 National Team (**Personal Best: 5.09s, 27 medals**).
- **Music** – ABRSM Piano Grade 7 and ABRSM Music Theory Grade 8
- **International Maker Award** - NRC 2020 Design and built an IOT-based Quarantine Recorder to aid COVID-19 test collection.
- **ASMO 2019 Bronze Medalist, KMC Malaysia 2016 Bronze Medalist, National Maths Olympiad 2016 - 87th place**
- **Malaysia Representative in Robotics, 2016-2018, IYRC 2017 Open Maker Category World Champion**